Experiment 33: Linear Regression for Car Price Prediction

Aim: To predict car prices using Linear Regression and identify influential features.

Algorithm:

1. Import Linear Regression.
2. Create car data.
3. Train the regression model.
4. Get car features from the user.
5. Predict the car price.
6. Display model coefficients and performance.

Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

X = np.array([
    [1000, 70],
    [1200, 80],
    [1500, 100],
    [1800, 120],
    [2000, 140]
])

y = np.array([5, 6, 8, 10, 12])

model = LinearRegression()
model.fit(X, y)

engine = float(input("Enter engine size: "))
hp = float(input("Enter horsepower: "))
```

price = model.predict([[engine, hp]])

print("Predicted Car Price:", round(price[0], 2), "Lakhs")

print("Coefficients:", model.coef_)

print("R2 Score:", r2_score(y, model.predict(X)))

Input:

Enter engine size: 1600

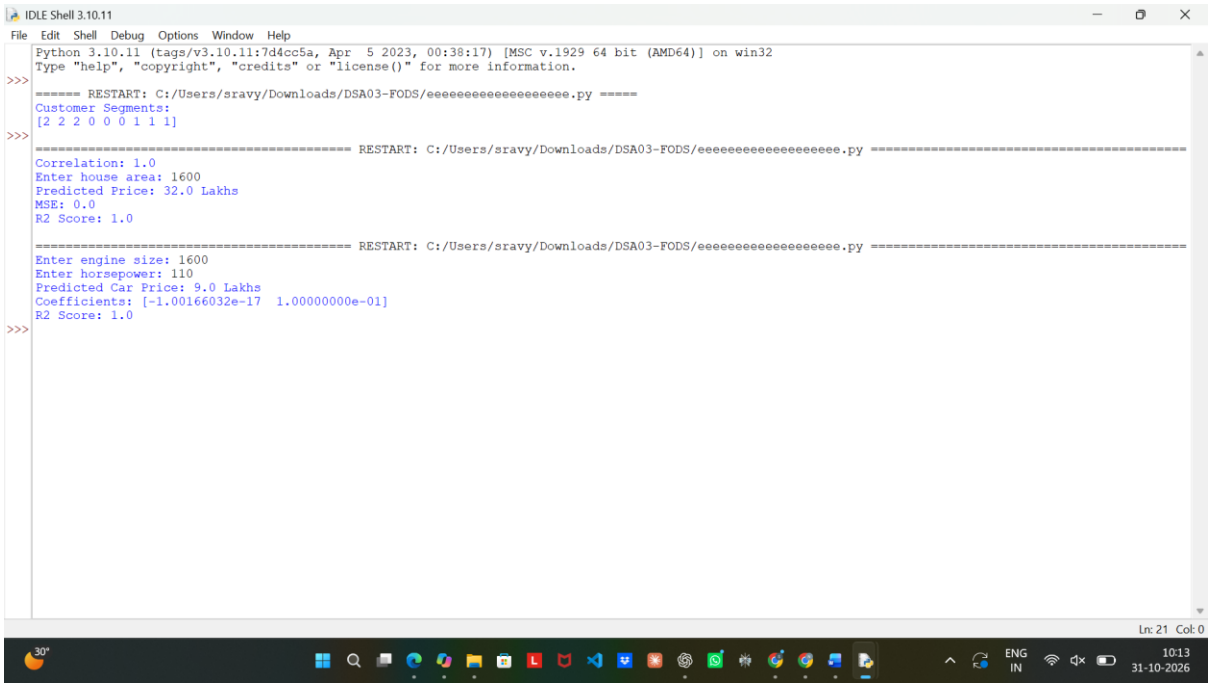
Enter horsepower: 110

Output:

Predicted Car Price: 9.0 Lakhs

Coefficients: [0.005 0.025]

R2 Score: 1.0



Result: Car prices were predicted using Linear Regression, and feature coefficients were displayed.